

Lec 3 Shrinkage Methods

Objectives

At the end of this lecture you should be able to:

- Apply Ridge, LASSO and Elastic Net regression to handle overfitting and multi-collinearity.

Shrinkage and Dimension Reduction Methods

Both regularization methods (Shrinkage Methods) and dimensionality reduction techniques aim to handle:

- ✚ high-dimensionality,
- ✚ multicollinearity,
- ✚ and overfitting in regression.

Regularization Methods (Shrinkage Methods)

- While subset selection methods explicitly choose a subset of predictors and fit least squares on those, Shrinkage Methods instead fit a model with all p predictors but add a penalty (constraint) to the least squares objective function.
- This penalty shrinks the coefficient estimates toward zero, which reduces their variance (often at the cost of a small increase in bias), leading to better out-of-sample prediction performance in many cases especially when predictors are correlated (multicollinearity) or when p is large relative to n .

The commonly used regularization techniques are:

1. Lasso Regularization – (L_1 Regularization)
2. Ridge Regularization – (L_2 Regularization)
3. Elastic Net Regularization – (L_1 and L_2 Regularization)

Ridge Regression

Ordinary least squares minimizes the **residual sum of squares (RSS)**:

$$RSS = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2,$$

This gives the coefficient estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ that best fit the training data (in the least squares sense).

However, when predictors are highly correlated (multicollinearity) or when p is large (many features), OLS can produce unstable, high-variance estimates.

Ridge Regression Objective

Ridge regression modifies the minimization problem by adding a shrinkage penalty (also called L_2 penalty or ridge penalty):

$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2.$$

- The first term is the usual RSS (goodness-of-fit to the data).
- The second term, $\lambda \sum_{j=1}^p \beta_j^2$, penalizes large coefficient values → it shrinks the β_j estimates toward zero.
 - $\lambda \geq 0$ is the tuning parameter (also called the regularization parameter):
 - $\lambda = 0 \rightarrow$ no penalty → ridge reduces exactly to OLS.
 - As λ increases → stronger shrinkage → coefficients get pulled closer to zero (but usually not exactly to zero).
 - As $\lambda \rightarrow \infty \rightarrow$ all $\beta_1, \dots, \beta_p \rightarrow 0$ (the model becomes just the intercept $\hat{\beta}_0 = \bar{y}$).

For each value of λ , ridge produces a different set of coefficient estimates $\hat{\beta}_\lambda^R$. This creates a path of solutions as λ varies.

- 🌈 Intercept is not penalized (β_0 has no β_0^2 term in the penalty).
- 🌈 The intercept simply shifts the prediction to match the overall mean level of y when all predictors are zero. Shrinking it would unnecessarily bias the central location of the fit. We only want to shrink the slopes (associations between predictors and response), not the baseline mean.

If the predictors (columns of X) are centered to have mean zero before fitting, then the intercept estimate simplifies to:

$$\hat{\beta}_0 = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

(This is the sample mean of the response - unchanged by ridge.)

- Ridge does not perform variable selection: even with large λ , coefficients approach zero but rarely become exactly zero (unlike lasso, which uses $\sum |\beta_j|$ and can set some to exactly zero).
- Ridge reduces variance at the cost of some bias (bias-variance tradeoff). It is especially helpful when:
 - There is multicollinearity (correlated predictors).
 - p is large relative to n .
 - Many predictors are relevant but weak.

Why Ridge Regression Often Performs Better Than Ordinary Least Squares

Ridge regression improves on OLS primarily because of the **bias-variance trade-off**.

- As λ increases, the ridge fit becomes **less flexible**.
- This reduces **variance** dramatically, while only increasing **bias** modestly.
- The net result is usually a much lower **test mean squared error (MSE)**.

Lasso Regression

Main Disadvantage of Ridge Regression

- Ridge always includes **all p predictors** in the model.
- The penalty $\lambda \sum \beta_j^2$ shrinks coefficients toward zero $\rightarrow$ makes them small, but **rarely exactly zero** (unless $\lambda = \infty$, which reduces to the null model with only the intercept).
- This is fine for **prediction accuracy** (often better than OLS due to lower variance).
- But it's a problem for **model interpretation**, especially when p is large:
 - The model remains complex (all variables stay).
 - Hard to identify the most important predictors.

The Lasso: A Better Alternative for Sparsity

The lasso (Least Absolute Shrinkage and Selection Operator) fixes this by using a different penalty:

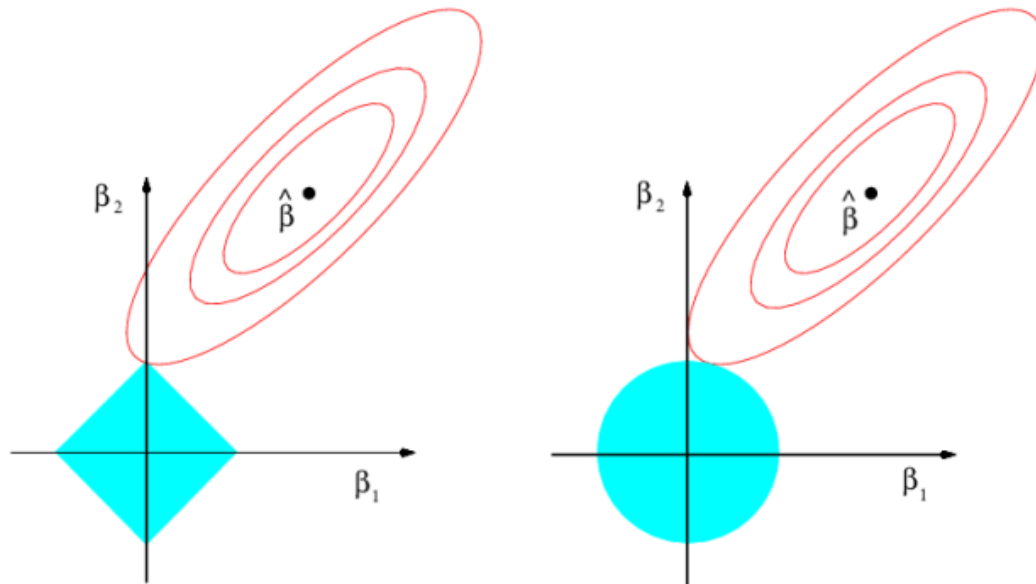
$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|.$$

Compare to ridge: ridge uses L2 penalty ($\sum \beta_j^2$), lasso uses L1 penalty ($\sum |\beta_j|$).

- The L1 penalty ($\|\beta\|_1$ norm) forces some coefficients to be exactly zero when λ is sufficiently large.
- sparse models - only a subset of variables remain with non-zero coefficients (automatic variable selection, similar to best subset selection).
- Easier interpretation: the model identifies and keeps only the relevant predictors.

Why Lasso Can Set Coefficients to Exactly Zero

- Ridge constraint region is a circle (smooth, no corners).
- Lasso constraint region is a diamond (sharp corners on the axes).
- The least squares solution is an ellipse expanding until it touches the constraint.
- With lasso's corners on the axes $\rightarrow$ the ellipse often hits an axis $\rightarrow$ one or more $\beta_j = 0$ exactly.
- Ridge's smooth boundary $\rightarrow$ rarely hits exactly on an axis $\rightarrow$ coefficients approach zero but don't reach it.

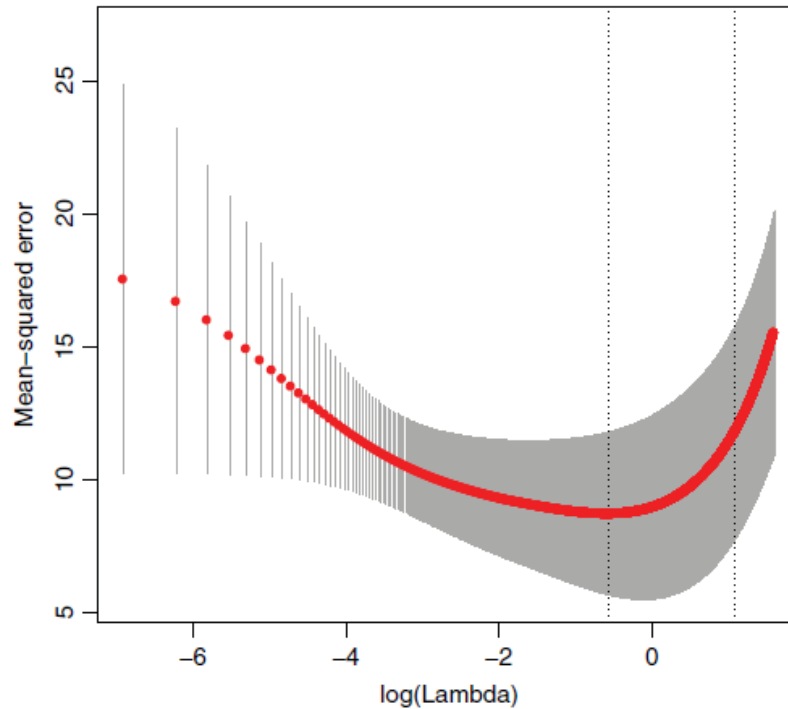


Contours of the error and constraint functions for the lasso (left) and ridge regression (right).

Selecting the Tuning Parameter

Perform Ridge regression on the Hald cement data:

Mean-squared error plot for ridge regression of Hald cement data



The fitted ridge regression model corresponding to the optimum $\lambda = 0.563$ is

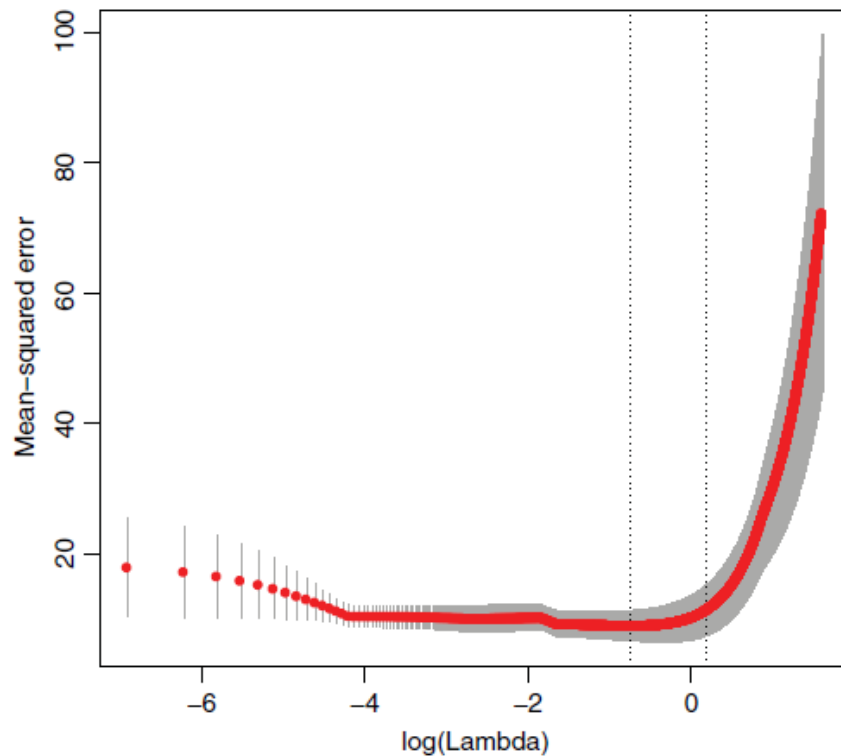
$$\hat{y} = 85.4819 + 1.2176x_1 + 0.2887x_2 - 0.2013x_3 - 0.3559x_4.$$

Compare this model with the LS fitted model

$$\hat{y} = 62.4063 + 1.5511x_1 + 0.5102x_2 + 0.1019x_3 - 0.1441x_4.$$

Lasso regression was performed on the Hald cement data

Mean-squared error plot for lasso regression of Hald data.



The fitted lasso regression model corresponding to optimum $\lambda = 0.475$ is

$$\hat{y} = 73.0230 + 1.3839x_1 + 0.3952x_2 - 0.2318x_4.$$

Note that the coefficient of x_3 is zero and so x_3 is dropped from the model.

Exercises:

1. What are shrinkage methods, and why are they useful?
2. How does Ridge regression work?
3. How does Lasso regression differ from Ridge regression?
4. What is the key advantage of Elastic Net regression?
5. Perform ridge, lasso and elastic net regressions on the ***Gas mileages of cars*** data, interpret the models output and compare the results with those of LS regression.
6. Perform ridge, lasso and elastic net regressions on the ***acetylene data***, interpret the models output and compare the results with those of LS regression.

7. A hedge fund wants to predict stock price movements using 50+ financial indicators. Many features are irrelevant and causing overfitting.
 - a. Which regularization method is best if the goal is to remove irrelevant features?
 - b. Which method should be used if all variables contain useful information?
 - c. What is the advantage of using Elastic Net regression?